WEAKLY ASSOCIATED PRIMES OF A TENSOR PRODUCT WITH A FLAT FINITELY PRESENTED MODULE OVER A FINITELY PRESENTED RING MAP

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Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The preliminary choice of the element g and the etale local model attached to a flat finitely presented module. Same chapter flat.tex, labels lemma-weak-bourbaki-pre-pre (invoked at proof line 3771) and proposition-finite-presentation-flat-at-point (invoked at proof line 3772). Chooses g and produces, after an etale base change R -> R' and S -> S', a situation where N tensor_S S' is projective over R'; the key observation (equation-key-observation) is then deduced inside the counted span. The proposition is itself a separate candidate in the same pool and is consumed here only as a pointer.
- Weakly associated primes under etale base change and universality of injectivity. Same chapter flat.tex, labels lemma-etale-weak-assassin-up-down (proof line 3799) and lemma-invert-universally-injective (proof lines 3848 and 3878). The first transports WeakAss, Ass and the weakly associated primes of the fibre through the etale maps, which is the bookkeeping of the first part of the proof; the second converts the non membership of a chosen element in the relevant primes into universal injectivity of multiplication by it, which is used twice.
- Elementary facts on associated and weakly associated primes, projectivity over a local ring, and the auto-associated criterion for projective modules. Algebra, labels algebra-lemma-ass-weakly-ass (proof line 3809), algebra-lemma-weakly-ass-local (proof line 3832), algebra-lemma-ass-zero-divisors, algebra-lemma-finite-ass and algebra-lemma-silly (proof lines 3840-3842),

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algebra-lemma-etale-at-prime (proof line 3889), algebra-theorem-projective-free-over-local-ring (proof line 3899) and algebra-lemma-weakly-ass (proof line 3907); Morphisms, label morphisms-lemma-etale-over-field (proof line 3923); More on Algebra, labels more-algebra-definition-auto-ass (proof line 3958) and more-algebra-lemma-P-fPD-zero (proof line 3962). These supply the local characterisation of weakly associated primes, the existence of an element of a prime ideal that is not a zerodivisor on a given module, discreteness of the fibres of an etale morphism, freeness of a projective module over a local ring, and the criterion that a non injective map of projective modules over an auto-associated ring can be detected after tensoring with the residue field.

Source reference. Original lemma 058A: sources/stacks-chow/tags.txt, source lines 6773–6773. Navigation pointer for \ref{algebra-lemma-ass-weakly-ass} used inside the counted span (line 3809); the argument that consumes it is retained in the span.

Source reference. Original lemma 00LD: sources/stacks-chow/tags.txt, source lines 783–783. Navigation pointer for \ref{algebra-lemma-ass-zero-divisors} used inside the counted span (line 3840); the argument that consumes it is retained in the span.

Source reference. Original lemma 00U4: sources/stacks-chow/tags.txt, source lines 1098–1098. Navigation pointer for \ref{algebra-lemma-etale-at-prime} used inside the counted span (line 3889); the argument that consumes it is retained in the span.

Source reference. Original lemma 00LC: sources/stacks-chow/tags.txt, source lines 782–782. Navigation pointer for \ref{algebra-lemma-finite-ass} used inside the counted span (line 3841); the argument that consumes it is retained in the span.

Source reference. Original lemma 00DS: sources/stacks-chow/tags.txt, source lines 508–508. Navigation pointer for \ref{algebra-lemma-silly} used inside the counted span (line 3842); the argument that consumes it is retained in the span.

Source reference. Original lemma 0548: sources/stacks-chow/tags.txt, source lines 6631–6631. Navigation pointer for \ref{algebra-lemma-weakly-ass} used inside the counted span (line 3907); the argument that consumes it is retained in the span.

Source reference. Original lemma 0566: sources/stacks-chow/tags.txt, source lines 6699–6699. Navigation pointer for \ref{algebra-lemma-weakly-ass-local} used inside the counted span (line 3832); the argument that consumes it is retained in the span.

Source reference. Original lemma 0593: sources/stacks-chow/tags.txt, source lines 6801–6801. Navigation pointer for \ref{algebra-theorem-projective-free-over-local-ring} used inside the counted span (line 3899); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FP: sources/stacks-flat/pinned-original-source.tex, source lines 353–353. Navigation pointer for \ref{lemma-etale-weak-assassin-up-down} used inside the counted span (line 3799); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FR: sources/stacks-flat/pinned-original-source.tex, source lines 1976–1976. Navigation pointer for \ref{lemma-invert-universally-injective} used inside the counted span (line 3848); the argument that consumes it is retained in the span.

Source reference. Original lemma 0CU6: sources/stacks-flat/pinned-original-source.tex, source lines 3705–3705. Navigation pointer for \ref{lemma-weak-bourbaki-pre-pre}

used inside the counted span (line 3771); the argument that consumes it is retained in the span.

Source reference. Original lemma 05GM: sources/stacks-chow/tags.txt, source lines 7066–7066. Navigation pointer for \ref{more-algebra-definition-auto-ass} used inside the counted span (line 3958); the argument that consumes it is retained in the span.

Source reference. Original lemma 05GQ: sources/stacks-chow/tags.txt, source lines 7069–7069. Navigation pointer for \ref{more-algebra-lemma-P-fPD-zero} used inside the counted span (line 3962); the argument that consumes it is retained in the span.

Source reference. Original lemma 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Navigation pointer for \ref{morphisms-lemma-etale-over-field} used inside the counted span (line 3923); the argument that consumes it is retained in the span.

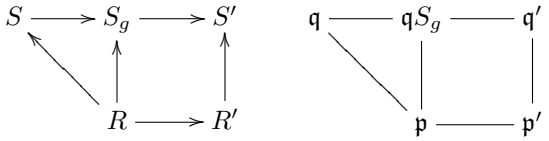
Source reference. Original lemma 05ID: sources/stacks-flat/pinned-original-source.tex, source lines 3287–3287. Navigation pointer for \ref{proposition-finite-presentation-flat-at-point} used inside the counted span (line 3772); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map of finite presentation. Let N be a finitely presented S-module which is flat as an R-module. Let M be an R-module. Let $\mathfrak{q}$ be a prime of S lying over $\mathfrak{p} \subset R$. Then*

$$\mathfrak{q} \in \mathit{WeakAss}_S(M \otimes_R N) \Leftrightarrow \Big(\mathfrak{p} \in \mathit{WeakAss}_R(M) \text{ and } \overline{\mathfrak{q}} \in \mathit{Ass}_{\overline{S}}(\overline{N})\Big)$$

Here $\overline{S} = S \otimes_R \kappa(\mathfrak{p})$, $\overline{\mathfrak{q}} = \mathfrak{q}\overline{S}$, and $\overline{N} = N \otimes_R \kappa(\mathfrak{p})$.

Proof. Pick $g \in S$ as in Lemma 0CU6. Apply Proposition 05ID to the morphism of schemes $\mathrm{Spec}(S_g) \to \mathrm{Spec}(R)$, the quasi-coherent module associated to N_g, and the points corresponding to the primes $\mathfrak{q}S_g$ and $\mathfrak{p}$. Translating into algebra we obtain a commutative diagram of rings



endowed with primes as shown, the horizontal arrows are étale, and $N \otimes_S S'$ is projective as an R'-module. Set $N' = N \otimes_S S'$, $M' = M \otimes_R R'$, $\overline{S}' = S' \otimes_{R'} \kappa(\mathfrak{q}')$, $\overline{\mathfrak{q}}' = \mathfrak{q}'\overline{S}'$, and

$$\overline{N}' = N' \otimes_{R'} \kappa(\mathfrak{p}') = \overline{N} \otimes_{\overline{S}} \overline{S}'$$

By Lemma 05FP we have

$$\mathrm{WeakAss}_{S'}(M' \otimes_{R'} N') = (\mathrm{Spec}(S') \to \mathrm{Spec}(S))^{-1}\mathrm{WeakAss}_S(M \otimes_R N)$$
$$\mathrm{WeakAss}_{R'}(M') = (\mathrm{Spec}(R') \to \mathrm{Spec}(R))^{-1}\mathrm{WeakAss}_R(M)$$
$$\mathrm{Ass}_{\overline{S}'}(\overline{N}') = (\mathrm{Spec}(\overline{S}') \to \mathrm{Spec}(\overline{S}))^{-1}\mathrm{Ass}_{\overline{S}}(\overline{N})$$

Use Algebra, Lemma 058A for $\overline{N}$ and $\overline{N}'$. In particular we have

$$\mathfrak{q} \in \mathrm{WeakAss}_S(M \otimes_R N) \Leftrightarrow \mathfrak{q}' \in \mathrm{WeakAss}_{S'}(M' \otimes_{R'} N')$$
$$\mathfrak{p} \in \mathrm{WeakAss}_R(M) \Leftrightarrow \mathfrak{p}' \in \mathrm{WeakAss}_{R'}(M')$$
$$\overline{\mathfrak{q}} \in \mathrm{Ass}_{\overline{S}}(\overline{N}) \Leftrightarrow \overline{\mathfrak{q}}' \in \mathrm{WeakAss}_{\overline{S}'}(\overline{N}')$$

Our careful choice of g and the formula for $\mathrm{Ass}_{\overline{S}'}(\overline{N}')$ above shows that

$$\text{if } \mathfrak{r}' \in \mathrm{Ass}_{\overline{S}'}(\overline{N}') \text{ lies over } \mathfrak{r} \subset \overline{S} \text{ then } \mathfrak{r} \subset \overline{\mathfrak{q}} \tag{0.1.1}$$

This will be a key observation later in the proof. We will use the characterization of weakly associated primes given in Algebra, Lemma 0566 without further mention.

Suppose that $\overline{\mathfrak{q}} \notin \mathrm{Ass}_{\overline{S}}(\overline{N})$. Then $\overline{\mathfrak{q}}' \notin \mathrm{Ass}_{\overline{S}'}(\overline{N}')$. By Algebra, Lemmas 00LD, 00LC, and 00DS there exists an element $\overline{a}' \in \overline{\mathfrak{q}}'$ which is not a zerodivisor on $\overline{N}'$. After replacing $\overline{a}'$ by $\lambda\overline{a}'$ for some nonzero $\lambda \in \kappa(\mathfrak{p})$ we can find $a' \in \mathfrak{q}'$ mapping to $\overline{a}'$. By Lemma 05FR the map $a' : N'_{\mathfrak{p}'} \to N'_{\mathfrak{p}'}$ is $R'_{\mathfrak{p}'}$-universally injective. In particular we see that $a' : M' \otimes_{R'} N' \to M' \otimes_{R'} N'$ is injective after localizing at $\mathfrak{p}'$ and hence after localizing at $\mathfrak{q}'$. Clearly this implies that $\mathfrak{q}' \notin \mathrm{WeakAss}_{S'}(M' \otimes_{R'} N')$. We conclude that $\mathfrak{q} \in \mathrm{WeakAss}_S(M \otimes_R N)$ implies $\overline{\mathfrak{q}} \in \mathrm{Ass}_{\overline{S}}(\overline{N})$.

Assume $\mathfrak{q} \in \mathrm{WeakAss}_S(M \otimes_R N)$. We want to show $\mathfrak{p} \in \mathrm{WeakAss}_S(M)$. Let $z \in M \otimes_R N$ be an element such that $\mathfrak{q}$ is minimal over $J = \mathrm{Ann}_S(z)$. Let $f_i \in \mathfrak{p}$, $i \in I$ be a set of generators of the ideal $\mathfrak{p}$. Since $\mathfrak{q}$ lies over $\mathfrak{p}$, for every i we can choose an $n_i \geq 1$ and $g_i \in S$, $g_i \notin \mathfrak{q}$ with $g_i f_i^{n_i} \in J$, i.e., $g_i f_i^{n_i} z = 0$. Let $z' \in (M' \otimes_{R'} N')_{\mathfrak{p}'}$ be the image of z. Observe that z' is nonzero because z has nonzero image in $(M \otimes_R N)_{\mathfrak{q}}$ and because $S_{\mathfrak{q}} \to S'_{\mathfrak{q}'}$ is faithfully flat. We claim that $f_i^{n_i} z' = 0$.

Proof of the claim: Let $g'_i \in S'$ be the image of g_i. By the key observation (0.1.1) we find that the image $\overline{g}'_i \in \overline{S}'$ is not contained in $\mathfrak{r}'$ for any $\mathfrak{r}' \in \mathrm{Ass}_{\overline{S}'}(\overline{N})$. Hence by Lemma 05FR we see that $g'_i : N'_{\mathfrak{p}'} \to N'_{\mathfrak{p}'}$ is $R'_{\mathfrak{p}'}$-universally injective. In particular we see that $g'_i : M' \otimes_{R'} N' \to M' \otimes_{R'} N'$ is injective after localizating at $\mathfrak{p}'$. The claim follows because $g_i f_i^{n_i} z' = 0$.

Our claim shows that the annihilator of z' in $R'_{\mathfrak{p}'}$ contains the elements $f_i^{n_i}$. As $R \to R'$ is étale we have $\mathfrak{p}' R'_{\mathfrak{p}'} = \mathfrak{p} R'_{\mathfrak{p}'}$ by Algebra, Lemma 00U4. Hence the annihilator of z' in $R'_{\mathfrak{p}'}$ has radical equal to $\mathfrak{p}' R_{\mathfrak{p}'}$ (here we use z' is not zero). On the other hand

$$z' \in (M' \otimes_{R'} N')_{\mathfrak{p}'} = M'_{\mathfrak{p}'} \otimes_{R'_{\mathfrak{p}'}} N'_{\mathfrak{p}'}$$

The module $N'_{\mathfrak{p}'}$ is projective over the local ring $R'_{\mathfrak{p}'}$ and hence free (Algebra, Theorem 0593). Thus we can find a finite free direct summand $F' \subset N'_{\mathfrak{p}'}$ such that $z' \in M'_{\mathfrak{p}'} \otimes_{R'_{\mathfrak{p}'}} F'$. If F' has rank n, then we deduce that $\mathfrak{p}' R'_{\mathfrak{p}'} \in \mathrm{WeakAss}_{R'_{\mathfrak{p}'}}(M'^{\oplus n}_{\mathfrak{p}'})$. This implies $\mathfrak{p}' R'_{\mathfrak{p}'} \in \mathrm{WeakAss}(M'_{\mathfrak{p}'})$ for example by Algebra, Lemma 0548. Then $\mathfrak{p}' \in \mathrm{WeakAss}_{R'}(M')$ which in turn gives $\mathfrak{p} \in \mathrm{WeakAss}_R(M)$. This finishes the proof of the implication "$\Rightarrow$" of the equivalence of the lemma.

Assume that $\mathfrak{p} \in \mathrm{WeakAss}_R(M)$ and $\overline{\mathfrak{q}} \in \mathrm{Ass}_{\overline{S}}(\overline{N})$. We want to show that $\mathfrak{q}$ is weakly associated to $M \otimes_R N$. Note that $\overline{\mathfrak{q}}'$ is a maximal element of $\mathrm{Ass}_{\overline{S}'}(\overline{N}')$. This is a consequence of (0.1.1) and the fact that there are no inclusions among the primes of $\overline{S}'$ lying over $\overline{\mathfrak{q}}$ (as fibres of étale morphisms are discrete Morphisms, Lemma 02GL). Thus, after replacing $R, S, \mathfrak{p}, \mathfrak{q}, M, N$ by $R', S', \mathfrak{p}', \mathfrak{q}', M', N'$ we may assume, in addition to the assumptions of the lemma, that

(1) $\mathfrak{p} \in \mathrm{WeakAss}_R(M)$,
(2) $\overline{\mathfrak{q}} \in \mathrm{Ass}_{\overline{S}}(\overline{N})$,
(3) N is projective as an R-module, and
(4) $\overline{\mathfrak{q}}$ is maximal in $\mathrm{Ass}_{\overline{S}}(\overline{N})$.

There is one more reduction, namely, we may replace R, S, M, N by their localizations at $\mathfrak{p}$. This leads to one more condition, namely,

(5) R is a local ring with maximal ideal $\mathfrak{p}$.

We will finish by showing that (1) – (5) imply $\mathfrak{q} \in \text{WeakAss}(M \otimes_R N)$.

Since R is local and $\mathfrak{p} \in \text{WeakAss}_R(M)$ we can pick a $y \in M$ whose annihilator I has radical equal to $\mathfrak{p}$. Write $\overline{\mathfrak{q}} = (\overline{g}_1, \ldots, \overline{g}_n)$ for some $\overline{g}_i \in \overline{S}$. Choose $g_i \in S$ mapping to $\overline{g}_i$. Then $\mathfrak{q} = \mathfrak{p}S + g_1S + \ldots + g_nS$. Consider the map

$$\Psi : N/IN \longrightarrow (N/IN)^{\oplus n}, \quad z \longmapsto (g_1z, \ldots, g_nz).$$

This is a homomorphism of projective R/I-modules. The local ring R/I is auto-associated (More on Algebra, Definition 05GM) as $\mathfrak{p}/I$ is locally nilpotent. The map $\Psi \otimes \kappa(\mathfrak{p})$ is not injective, because $\overline{\mathfrak{q}} \in \text{Ass}_{\overline{S}}(\overline{N})$. Hence More on Algebra, Lemma 05GQ implies Ψ is not injective. Pick $z \in N/IN$ nonzero in the kernel of Ψ. The annihilator $J = \text{Ann}_S(z)$ contains IS and g_i by construction. Thus $\sqrt{J} \subset S$ contains $\mathfrak{q}$. Let $\mathfrak{s} \subset S$ be a prime minimal over J. Then $\mathfrak{q} \subset \mathfrak{s}$, $\mathfrak{s}$ lies over $\mathfrak{p}$, and $\mathfrak{s} \in \text{WeakAss}_S(N/IN)$. The last fact by definition of weakly associated primes. Apply the "$\Rightarrow$" part of the lemma (which we've already proven) to the ring map $R \to S$ and the modules R/I and N to conclude that $\overline{\mathfrak{s}} \in \text{Ass}_{\overline{S}}(\overline{N})$. Since $\overline{\mathfrak{q}} \subset \overline{\mathfrak{s}}$ the maximality of $\overline{\mathfrak{q}}$, see condition (4) above, implies that $\overline{\mathfrak{q}} = \overline{\mathfrak{s}}$. This shows that $\mathfrak{q} = \mathfrak{s}$ and we conlude what we want. $\square$